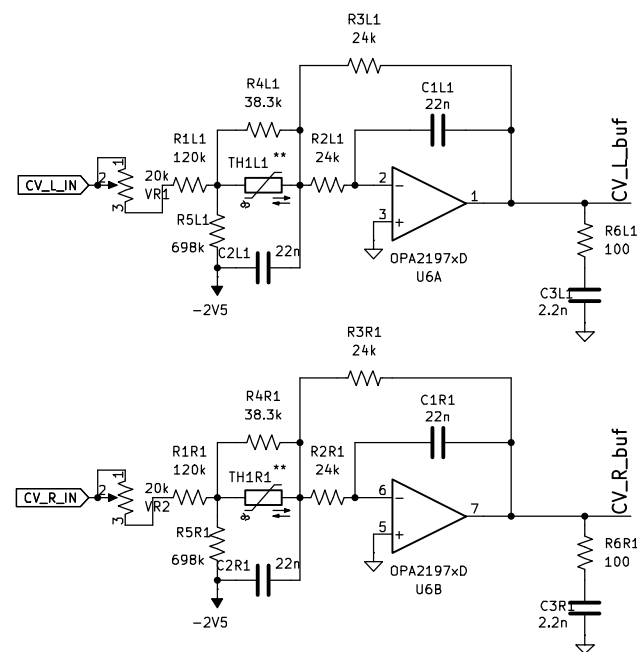


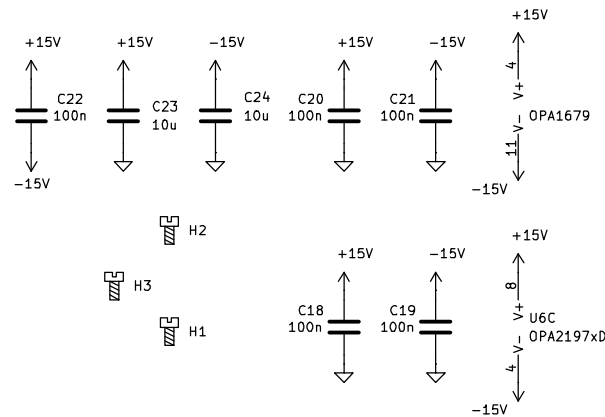
CONTROL VOLTAGE BUFFERS



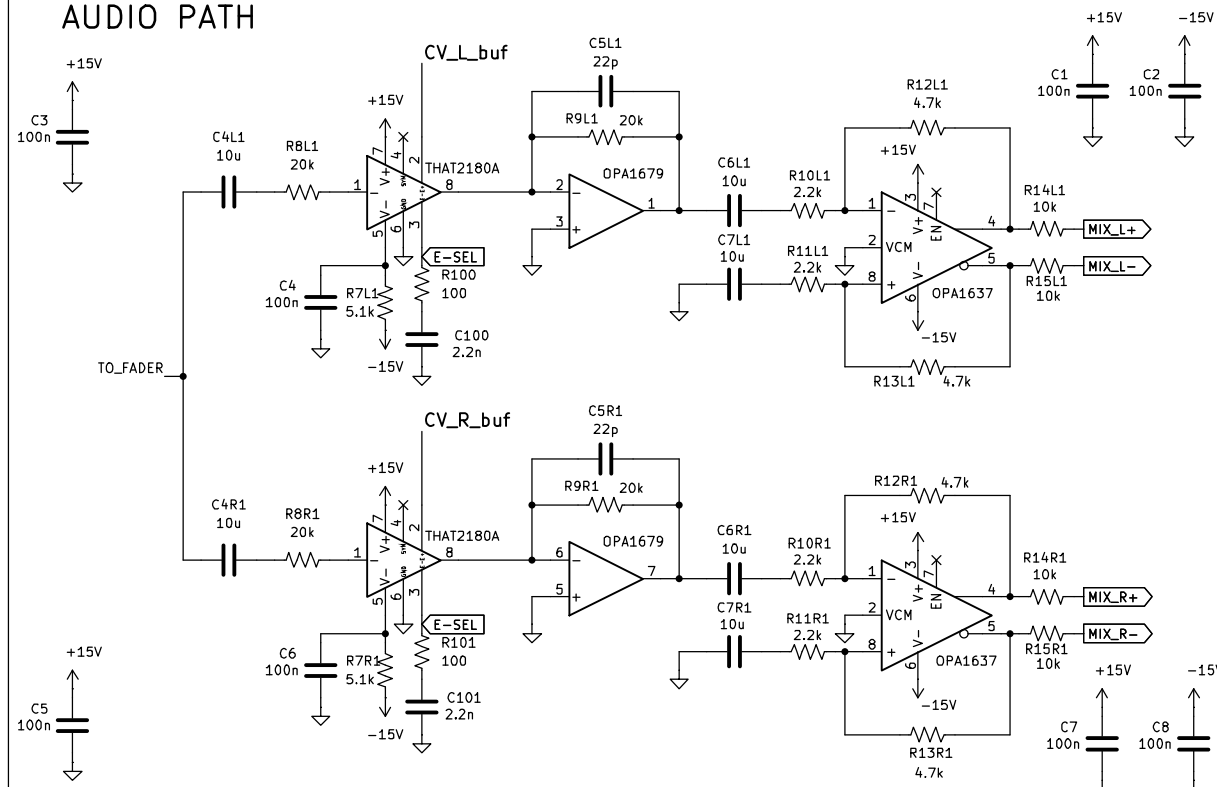
CV Buffer Functions:

1. Buffer to drive the VCA CV input
2. LPF at ~301Hz, multi-feedback two pole
3. Rescale 3.3V range to 0.5V range (Gives 82dB of gain control range)
4. Shift range to 73mV to -427mV (73mV = +12dB, -427mV = -70dB)
5. Correct temperature drift by scaling CV to compensate (using NTC thermistor)

** Neg. Temp. Coeff. (NTC) Thermistor
Part No. NTCLE100E3473JB0 Vishay

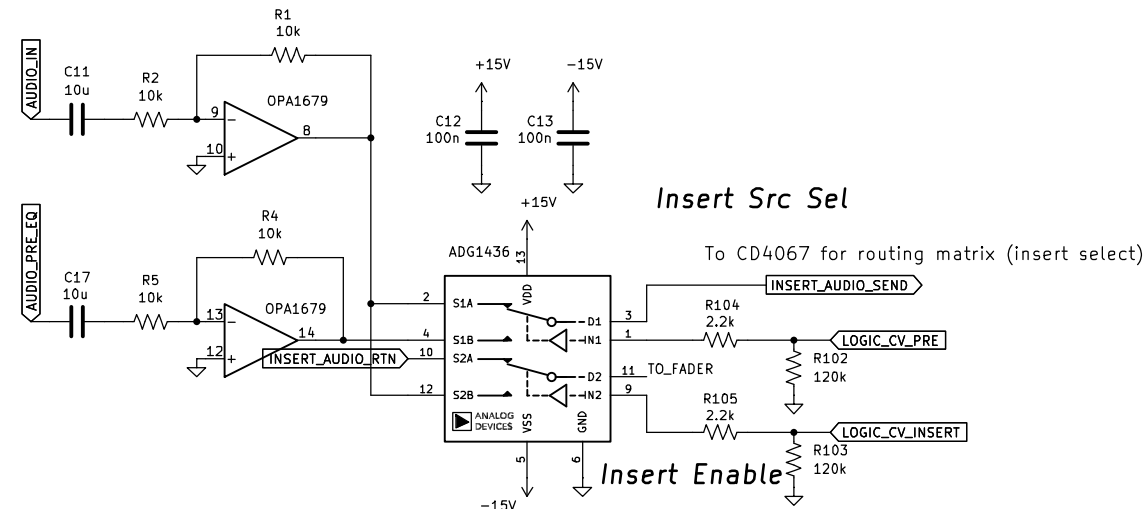


AUDIO PATH



CHANNEL INSERT SWITCH NETWORK

Insert Source Buffers



DESIGN NOTES

- Use film capacitors for all in audio path, <= 5%
- Use THIN film resistors for all in audio path, <= 0.5%
- R6 and C3 should physically close to CV input pin 2 of VCAs
- The NTC thermistor (TH1) has a nominal resistance 47k (@25C)

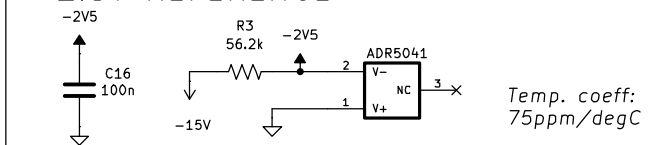
CV BUFFER DESIGN:

- CV LPF cutoff = $1/(R2 \cdot C1 \cdot 2\pi) = 1/(2\pi \cdot 22n \cdot 24k) = 301Hz$
- CV range rescaling (gain factor) from 3.3V to 0.5V:
 $A = -R3 / (R1 + (R4 || TH1)) = -24k / (133k + (38.3k || 47k))$
 $= -0.151515 = -0.5V / 3.3V$
- The -2.5V reference is to offset the CV while rescaling it.
- The raw CV is in range 0-3.3V
- Just scaling it to range of 0.5V would give 0 to -0.5V, since configuration is inverting
- But we need to move up by 73mV to get -427mV to +73mV.
- $(-2.5V) / (698k + (133k || 38.3k || 47k)) = -3.49uA$
And the fraction reaching the summing node is
 $133k / (133k + (38.3k || 47k)) = 0.863$
Then, $-1 \cdot -3.49uA \cdot 24k \cdot 0.863 = 73mV$ (-1 since inverting)
By superposition, that gets added to the scaled voltage to move it up 73mV, bringing it into the right range.

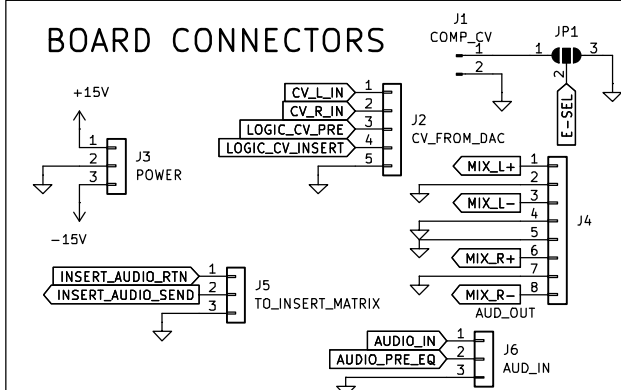
TEMPERATURE COMPENSATION

- The VCA drifts by 0.33% of the gain per deg. C.
- I.e. at -20dB gain, chip warming by 15C gives
 $-20dB \cdot 0.33\% \cdot 15C = +0.99dB \Rightarrow$ now -19.01dB
- To compensate CV scaling, need $0.33\% \cdot 133k = +439R$ / degC, make CV smaller, to cancel temp.
- This 47k NTC thermistor does 447R per C (close)
- Paralleling it with R4 makes it more linear (by default NTC is exponential, R4 helps extremes)

-2.5V REFERENCE



BOARD CONNECTORS



Fader, Pan, Channel Output

Analog Board

REV:
2

PROJECT: PAN_FADER	DRAWN: S.E.D.	SHEET: / FILE: PAN_FADER.kicad_sch
EECE CAPSTONE	CHECKED:	DATE: 2026-10-05